

SDG GreenTech Project

BCA411-7

Progress Report

BY

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SUBMITTED TO

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PALADINMEDIA ENGINEERING STATUS AUDIT

Project Status and Completion Report

Detailed Engineering Status, Implementation Verification, and Milestone Completion Audit

Project Name	PaladinMedia	**Current Baseline**	Foundation MVP 0 (Phase 2)
Automated Tests	99 / 99 Passed (100%)	**Sprint 1 Checks**	24 / 24 Passed (100%)
Report Date	October 2026	**Software Revision**	v0.2.0 (Phase 2 Baseline)
Core Completion	100% (Phases 0–2)	**Overall Progress**	~45% (Full 6-Phase Scope)

1. Executive Summary & Project Objectives

PaladinMedia is an advanced media security platform designed to conceal sensitive digital payloads (text, JSON configurations, high-entropy binaries, and raster images) inside benign-looking 512×512 RGB carrier images. The system solves the vulnerability of conventional encryption—which alerts network monitors by exposing high-entropy ciphertext—while simultaneously overcoming the fragility and lack of cryptographic security inherent in traditional steganography.

The project enforces a core architectural rule: "Security Before Concealment". Sensitive content is not merely hidden; it is first framed with canonical wire headers and integrity checksums, encrypted using authenticated AES-256-GCM, fortified against channel noise using systematic Reed-Solomon GF(2^8) error correction, and only then embedded into the discrete wavelet domain (2D Haar DWT) of a carrier image. On the receiver side, a strictly symmetric extraction pipeline performs error correction, authenticates the cryptographic tag, and recovers the exact original payload with 100% byte fidelity.

2. Current Implementation Status

The software architecture has been successfully engineered across three primary operational subsystems: the Secret Cryptographic Pipeline, the Transform-Domain Carrier Pipeline, and the Receiver Verification Pipeline.

2.1 Cryptographic & Security Pipeline (Phase 1 Baseline)

• Secret Serialization (src/core/serialization/serializer.py): Encapsulates payloads into canonical wire frames with magic bytes b'PLDS', format versioning, media type identifiers, payload length fields, and CRC-32 integrity trailers.
• Authenticated Cryptography (src/security/crypto/provider.py): Utilizes AES-256-GCM authenticated cipher with 96-bit CSPRNG nonces, 128-bit GHASH authentication tags, and transcript-bound Additional Authenticated Data (AAD). Guarantees zero plaintext leakage upon tampering.
• Galois Field Error Correction (src/security/ecc/encoder.py): Applies systematic Reed-Solomon GF(2^8) chunking with 2t=32 parity symbols per 255-byte block. Successfully corrects up to t=16 symbol errors (128 contiguous flipped bits) per block.

2.2 Carrier Signal Processing & Embedding Pipeline (Phase 2 Baseline)

• 2D Wavelet Decomposition (src/carrier/dwt.py): Performs one-level 2D Haar Discrete Wavelet Transform (DWT) in float64 precision, decomposing 512×512 RGB carrier images into LL, LH, HL, and HH subbands (3×256×256 shape) with zero numerical inversion loss.
• Capacity Accounting & Hard Gating (src/carrier/capacity.py): A 512×512 carrier offers 65,536 bits (8,192 bytes) of raw space in the Red channel of the HH subband. Reserving 16 bytes for the carrier header leaves 8,176 bytes maximum protected payload (~7,088 bytes of secret data after RS, AES, and wire overheads). Payloads exceeding this threshold are immediately rejected by a hard gate.
• Carrier Header & Deterministic Masking (src/carrier/header.py & mask.py): Injects a 16-byte fixed canonical header (b'PLDH') into the first 128 positions. Computes pseudo-random payload positions across the HH subband using a session-derived PRNG seed.
• Wavelet Embedder & Extractor (src/carrier/embedding.py & extraction.py): Modulates integer parity on 2×HH coefficients with carrier pixel adjustments (±1 pixel), saving lossless 8-bit PNG stego containers that survive image roundtripping.

2.3 Application & Observability Infrastructure

• Backend Server & Event Bus (src/api/server.py & event_adapter.py): A lightweight FastAPI server exposing REST endpoints for conceal/extract actions and streaming server-sent events for pipeline telemetry.
• Laboratory Frontend UI (frontend/): A React + TypeScript + Vite interactive laboratory interface scaffolded with Tailwind CSS, ready for coefficient visualization and interactive steganography analysis.

3. Verification & Empirical Testing Results

The project has undergone rigorous automated testing across three independent evaluation harnesses:

Testing Area / Metric	Measured Value	Target Threshold	Verification Result
Automated Pytest Suite	**99 / 99 Passed (100%)**	100% Pass	PASSED: Full unit & integration coverage across all modules (2.38s).
Sprint 1 Invariant Verification	**24 / 24 Passed (100%)**	24 Checks	PASSED: Multi-media reversibility, bounds, and fail-closed security.
DWT Mathematical Invertibility	**0.000000 max error**	0.000000	PASSED: Float64 Haar transform exhibits zero numerical drift.
Peak Signal-to-Noise Ratio (PSNR)	**75.71 dB**	> 50.0 dB	PASSED: Carrier distortion is invisible; exceeds perceptual stealth bar.
Structural Similarity (SSIM)	**1.00000**	> 0.999	PASSED: Identical image structure preserved across all 3 color channels.
Pre-ECC Bit Error Rate (BER)	**0.000000**	< 0.01	PASSED: Zero bit flips introduced through PNG carrier storage and reload.
Reed-Solomon Fault Tolerance	**Repairs t=16 symbols**	t ≤ 16 symbols	PASSED: Corrects burst errors up to 128 consecutive flipped bits.
End-to-End Secret Recovery	**100.0% Byte Match**	Formal Equality	PASSED: original_secret.data ==

recovered_secret.data is strictly TRUE.

4. Project Completion Status & Roadmap

The PaladinMedia master architecture is structured into seven distinct phases (Phase 0 through Phase 6). The following completion matrix audits the exact delivery status of each phase:

Phase	Milestone & Scope	Target Baseline	Completion	Current Status & Deliverables
Phase 0	Sprint-0 Readiness & Specs	Governance Docs	**100%**	COMPLETE: Product Vision, Architecture, and IRS frozen.
Phase 1	Cryptographic & ECC Core	Sprint 1 Foundation	**100%**	COMPLETE: AES-256-GCM, RS GF(2^8), fail-closed security verified.
Phase 2	Signal Processing & DWT Carrier	Foundation MVP 0	**100%**	COMPLETE: Haar DWT/IDWT, mask, E03 verified (99 tests pass).
Phase 3	ACS-Sub-INN Neural Predictor	Context Predictor CNN	**0%**	PENDING: Architecture specified; awaiting training harness.
Phase 4	Model Training on DIV2K	800 DIV2K Images	**0%**	PENDING: DIV2K dataset staged in workspace; ready for training.
Phase 5	Edge Optimization & QAT	ONNX Opset 14 / INT8	**0%**	PENDING: Quantization-aware training after neural model convergence.
Phase 6	Laboratory GUI & Telemetry	FastAPI + Vite + Energy	**35%**	IN PROGRESS: FastAPI server operational; Vite UI scaffolded.

4.1 Completion Assessment Summary

• Core Foundational Architecture (MVP 0 — Phases 0, 1, and 2): 100% COMPLETE AND VERIFIED. The end-to-end sender-to-receiver pipeline (serialization → AES-256-GCM encryption → Reed-Solomon encoding → 2D Haar DWT → deterministic PRNG masking → carrier embedding → IDWT → lossless PNG storage → symmetric extraction → decryption → 100% exact byte recovery) is fully engineered, covered by 99 automated tests, and validated.
• Overall Research Scope (Phases 0 through 6): ~45% COMPLETE. While the core steganographic and cryptographic engines are production-ready and verified, advanced neural learning (ACS-Sub-INN), edge deployment (ONNX INT8 QAT), and carbon telemetry represent the remaining forward research scope.

5. Observed Boundaries, Limitations & Next Actions

• Lossless Container Boundary: The current baseline is validated across lossless PNG storage. It does not claim robustness against hostile lossy recompression (JPEG) or geometric scaling, which will be tackled during Phase 3 neural robustness studies.
• Capacity Limit: Current baseline is constrained to ~7,088 bytes of secret data per 512×512 carrier image in 1 HH subband. Oversized payloads are safely rejected by the hard capacity gate without corrupting carrier coefficients.
• Non-Claims: PaladinMedia is an academic research platform; it makes no claims of operational military certification or unconditional 'unbreakability'.

Immediate Next Actions:
1. Complete Vite UI connection to the FastAPI backend to provide live browser-based carrier inspection.
2. Add CodeCarbon energy telemetry to track Joules and carbon emissions per embedding run.
3. Launch Phase 3 PyTorch implementation of the ACS-Sub-INN context predictor on the DIV2K training set.